

Acknowledgments

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中文摘要

摘要

關鍵詞:



Abstract

Named Entity Recognition (NER) in new or specialized domains often requires manually defining entity types and annotating data, a process that is costly and demands domain expertise. This limitation motivates **open-world NER**, where the goal is not only to detect entity mentions from unlabeled text but also to **discover and name latent entity types** automatically. In this thesis, we study unsupervised open-world NER under domain shift, where entity types are not predefined and performance depends on both high-recall mention detection and discriminative entity representations for clustering. Building on the OWNER framework, we propose three enhancements. First, we replace OWNER’s mention detection with a multi-view divide-and-transfer framework (2025): a shared text encoder feeds three complementary decoding heads—BIO tagging, start/end boundary tagging, and tie/break span linkage—which are ensembled and filtered to produce robust, high-recall entity spans. Second, for entity typing, we introduce soft-prompt entity encoding by prepending trainable virtual tokens to a BERT-style encoder and representing each mention via the contextualized hidden state at the prompt’s [MASK] position, improving task conditioning for open-world clustering. Third, we propose a cluster naming framework to generate human-readable entity type names for a given corpus: we select diverse cluster exemplars via maximal marginal relevance (MMR) and infer concise category names using either masked language modeling (MLM) consensus or a local LLM (Ollama), and evaluate naming quality with BERTScore. Together, these components form an end-to-end, more robust and interpretable unsupervised

open-world NER system that not only clusters entities but also produces readable type inventories for new corpora.

Keywords: Open-world Named Entity Recognition; Unsupervised NER; Domain Shift; Mention Detection; Multi-view Decoding; Entity Typing; Soft Prompting; Representation Learning; Clustering; Cluster Naming; Masked Language Modeling; Local LLM



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Chapter 1

Introduction

Named Entity Recognition (NER) is a foundational task in natural language processing (NLP), underpinning a wide range of downstream applications including information extraction, knowledge graph construction, question answering, biomedical text mining, and document understanding. The objective of NER is to locate mentions of named entities in text and classify them into predefined semantic categories such as PERSON, ORGANIZATION, or LOCATION. Over the past decade, advances in deep learning and pretrained language models have pushed NER performance to near-human levels on standard benchmarks such as CoNLL-2003 [1] and OntoNotes [2].

Despite this progress, virtually all high-performing NER systems operate under a *closed-world assumption*: the set of entity types is fixed and known in advance, and models are trained on large amounts of manually annotated data conforming to that type inventory. This assumption, while convenient for benchmark evaluation, creates a fundamental barrier to deploying NER in practice. In real-world settings, new domains constantly emerge—scientific subfields, legal jurisdictions, industrial verticals, social media communities—each with its own vocabulary of relevant entities that may bear little resemblance to the coarse types defined in general-domain corpora. Annotat-

ing sufficient labeled data for every new domain is expensive, slow, and often requires specialized expert knowledge. Furthermore, even within a single domain, entity categories are not static: terminology evolves, new entity types emerge, and the granularity of the desired ontology varies by application.

These limitations motivate a shift toward *open-world NER*: a paradigm in which the system receives an unannotated corpus and is expected to (1) identify entity mention spans, (2) group those spans into coherent semantic categories without being told how many categories exist or what they are called, and (3) produce human-readable names for the discovered categories. The third requirement—*automatic entity type naming*—is often treated as an afterthought in existing work, yet it is critical for practical deployment. Clusters of entity mentions without interpretable labels cannot be validated by domain experts, integrated into knowledge bases, or used as supervision signals for downstream tasks. Naming is therefore not merely a cosmetic concern; it is what transforms unsupervised cluster assignments into a usable, inspectable entity type ontology.

This thesis presents **Span2Type**, an improved open-world NER pipeline built on top of the OWNER framework [3], with two targeted enhancements. Span2Type operates in three stages. First, the mention detection module from OWNER—which uses a single BIO sequence labeling head—is replaced with the Entity Detection module from the Divide-and-Transfer (DTrans) paradigm [4]. The DTrans module uses BERT with three complementary tagging schemes applied simultaneously: standard BIO tagging for global span context, Start-End (SE) binary classifiers for explicit boundary detec-

tion, and Tie-Break (TB) classifiers that model whether adjacent tokens belong to the same span. Training all three views jointly with a Mean Teaching strategy (exponential moving average) improves domain transfer robustness, since entity span boundaries are more stable across domains than entity types. Second, the entity typing module encodes each detected span using the fill-in-the-blank template “*{sentence} {entity} is a [MASK].*” and extracts the [MASK]-position hidden state from a BERT-based encoder. This thesis introduces a soft prompt variant of this stage: a small number of trainable soft token embeddings are prepended to the template input while the base encoder is kept frozen, so that only the prompt parameters (not the full model) are updated during triplet margin training. The resulting embeddings are clustered with automatic k selection via the Bayesian Information Criterion (BIC). Third, the cluster naming module—the main novel contribution of this thesis—selects a diverse, representative set of exemplar sentences for each cluster using Maximal Marginal Relevance (MMR) and applies either a masked language model or a locally hosted large language model (LLM) to generate a concise, human-readable type name by aggregating predictions over the selected exemplars.

Span2Type requires no predefined type inventory and no target-domain annotations. The mention detector is trained on a source-domain corpus (CoNLL-2003 or Pile-NER) and transferred to new domains without retraining. Experiments on CrossNER spanning five target domains demonstrate that replacing OWNER’s BIO-only mention detector with the DTrans multi-view entity detector improves span recall across domains, and that MMR-

based exemplar selection produces more specific and stable cluster names compared to OWNER’s original naming procedure.

The main contributions of this thesis are as follows:

- We integrate the DTrans multi-view Entity Detection module [4] into the OWNER open-world NER pipeline, replacing its single BIO mention detector with a joint BIO + Start-End + Tie-Break decoder that yields higher span recall in cross-domain settings.
- We propose an automatic cluster naming procedure that uses Maximal Marginal Relevance [5] to select a diverse set of exemplar sentences per cluster before querying an MLM or LLM, improving naming quality and stability over the centroid-nearest baseline used in the original OWNER system.
- We provide a systematic empirical evaluation on CrossNER across five target domains, comparing the improved pipeline against the original OWNER baseline and reporting entity detection F1, entity typing AMI, and cluster naming quality via BERTScore.

1.1 Motivation

The development of Span2Type is driven by three concrete challenges that arise when attempting to deploy NER systems beyond well-curated benchmark settings.

Challenge 1: Robust span detection across domains without target supervision. The first stage of any open-world NER pipeline is detecting entity mention spans in the target corpus without access to target-domain labels. The standard approach, used in OWNER, is to train a BIO sequence tagger on a labeled source domain (e.g., CoNLL-2003) and apply it directly to the target domain. While effective in closely related domains, single-view BIO taggers struggle when target domain text differs substantially in structure or vocabulary from the source. Entity boundary signals—such as whether a token starts a new mention or is connected to the preceding token—can be captured from multiple complementary perspectives. A multi-view detector that jointly trains BIO, Start-End, and Tie-Break decoders can exploit these different signals simultaneously, producing span candidates that are more robust to domain shift. This thesis addresses this challenge by replacing OWNER’s BIO-only mention detector with the DTrans multi-view Entity Detection module [4].

Challenge 2: Unknown and evolving entity type inventories. Even in domains where NER data exist, the type inventory may be incomplete or misaligned with the application’s needs. Standard NER benchmarks use small, coarse-grained label sets (typically four to eighteen types). Many real-world applications require finer granularity—distinguishing, for example, between DRUG, CHEMICAL, and DISEASE rather than a single MEDICAL type—or operate on entity categories that have no counterpart in existing ontologies, such as entities specific to proprietary product catalogs or newly coined technical terms. Purely supervised methods cannot discover new types; they can only

classify entities into the types they were trained on. Open-world NER, by contrast, makes no assumptions about the number or nature of entity types and can surface latent categories that would be missed by any closed-world system.

Challenge 3: The interpretability gap in unsupervised entity clustering.

Existing unsupervised entity typing methods typically produce cluster assignments without associated type names. The output of such a system is a mapping from entity spans to cluster identifiers (e.g., *Cluster 3*, *Cluster 7*), which carries no semantic content. Practitioners who wish to use these assignments must manually inspect cluster members, guess an appropriate label, and verify that the label is consistent across the cluster—a process that reintroduces the very human effort that unsupervised methods were supposed to eliminate. Automatic type naming bridges this interpretability gap: by generating a concise label such as PHARMACEUTICAL COMPANY or PROTEIN COMPLEX directly from cluster evidence, the system enables human experts to validate the discovered ontology and integrate it into downstream pipelines with minimal manual intervention.

Together, these three challenges motivate the two improvements this thesis makes to the OWNER pipeline. The DTrans multi-view entity detector addresses Challenge 1 by capturing complementary span boundary signals that transfer better across domains. The MMR-based cluster naming module addresses Challenge 3 by producing human-readable type names from diverse cluster evidence rather than leaving practitioners with anonymous

cluster identifiers. Challenge 2—the unknown number of entity types—is handled by the BIC-guided k selection already present in OWNER, which this thesis retains unchanged.

1.2 Contributions

This thesis makes the following specific technical contributions:

- **Multi-view entity detection for open-world NER.** We replace the BIO-only mention detector in OWNER with the DTrans multi-view Entity Detection module [4], which jointly trains three complementary decoders—BIO, Start-End, and Tie-Break—on the source domain. The Mean Teaching strategy (exponential moving average) further stabilizes training, yielding more reliable span candidates when transferred to unseen target domains.
- **Soft prompt tuning for entity typing.** We introduce a parameter-efficient variant of the entity typing stage based on soft prompt tuning. Rather than fine-tuning all parameters of the BERT encoder, a small set of trainable soft tokens is prepended to the fill-in-the-blank input template while the base language model is kept frozen. Only the soft prompt embeddings (and optionally a separate low learning rate for the PLM) are optimized via a triplet margin loss, significantly reducing the number of trainable parameters while preserving the quality of the [MASK]-position entity representations used for clustering.
- **MMR-based exemplar selection for automatic cluster naming.**

We propose selecting a diverse, centroid-relevant set of exemplar sentences per cluster using Maximal Marginal Relevance [5] before querying either an MLM or a locally hosted LLM for type name generation. Compared to OWNER’s original naming procedure, MMR-based selection reduces exemplar redundancy and produces more specific and stable cluster names.

- **Empirical evaluation on CrossNER.** We evaluate the improved pipeline on five CrossNER target domains [6], reporting span detection F1, entity typing Adjusted Mutual Information (AMI) and Macro-F1, and cluster naming quality via BERTScore, with ablations that isolate the contribution of each component.

Chapter 2

Related Work

2.0.1 Supervised Named Entity Recognition

Named Entity Recognition has a long history as a sequence labeling problem. Early systems relied on hand-crafted features and probabilistic models such as Hidden Markov Models and Conditional Random Fields (CRFs) [7]. The introduction of bidirectional LSTM-CRF architectures [8] significantly improved performance by learning contextual token representations without manual feature engineering. The transformer architecture [9] and pretrained models such as BERT [10] and RoBERTa [11] have become the de facto backbone for NER, achieving state-of-the-art results on benchmarks including CoNLL-2003 [1] and OntoNotes [2]. Span-based models [12] offer an alternative to token-level BIO labeling by directly enumerating and classifying candidate spans. All of these approaches share a common limitation: they require annotated data for a fixed, predefined set of entity types and cannot generalize to type inventories unseen during training.

2.0.2 Cross-domain and Low-resource NER

A line of work addresses the cost of annotation by transferring NER models across domains or reducing the amount of supervision required. Cross-domain NER studies explore domain adaptation techniques, multi-task learning, and data augmentation to transfer models trained on newswire to specialized domains such as biomedical or social media text [13]. Few-shot NER approaches [14, 15] use prototypical networks or meta-learning to recognize new entity types from a handful of examples. While these methods reduce the annotation burden, they still assume that the target type inventory is specified in advance and that at least a small number of labeled examples per type is available. In contrast, this thesis targets the strictly unsupervised setting where no target-domain labels of any kind are provided.

2.0.3 Open-world and Unsupervised Entity Typing

Open-world NER removes the closed-world assumption entirely: the system must discover entity types from raw text without any predefined ontology or labeled examples. Representative work in this space embeds entity mentions using pretrained language models and applies clustering to group them by type. Fill-in-the-blank (cloze) templates have been proposed as a way to obtain type-sensitive embeddings by exploiting the knowledge stored in masked language models [16, 17]. More recent systems such as UniversalNER [18] and GLiNER [19] use instruction-tuned LLMs to perform open-vocabulary NER, while ChatIE [20] applies zero-shot prompting. The OWNER frame-

work [3] is a representative clustering-based pipeline: it trains a DeBERTa-based [21] mention detector on a source domain and clusters BERT [MASK]-position embeddings on the target domain using k -means with automatic k selection via the Bayesian Information Criterion (BIC). This thesis builds directly on OWNER and focuses on improving its span detection and cluster naming components.

2.0.4 Fine-grained and Ultra-fine Entity Typing

Fine-grained and ultra-fine entity typing [22, 23] assign more detailed semantic types to entity mentions, often using large type vocabularies derived from knowledge bases. While these approaches provide richer type information than coarse NER, they still rely on a predefined type ontology and treat entity typing as a supervised classification problem. They do not address the open-world challenge of discovering new entity types without any labeled data.

2.0.5 Clustering Methods for Entity Typing

Most embedding-based entity typing systems rely on k -means [24] or its variants for cluster assignment. The k -means++ initialization [25] significantly improves convergence stability over random initialization. A practical challenge is that the correct number of clusters k is unknown in the open-world setting. The Bayesian Information Criterion [26] penalizes model complexity and provides a principled trade-off between fit quality and par-

simony. Clustering quality is measured using the Adjusted Mutual Information (AMI) [27], a metric robust to over-segmentation and over-merging. This thesis uses BIC-guided k -means as implemented in OWNER, treating cluster selection as a fixed component and concentrating improvements on detection and naming.

2.0.6 Masked Language Model Probing for Type Inference

Masked language models such as BERT have been widely used as knowledge bases that can be queried via cloze prompts to infer semantic properties of entities [16]. For entity typing, the standard approach is to construct a template such as “*{sentence} {entity} is a [MASK]*” and take the top-ranked token at the mask position as the predicted type [17]. The quality of this prediction is sensitive to prompt design: different templates yield different tokens, and the top-ranked token is often a generic or uninformative word. Aggregating predictions over multiple instances of the same cluster, rather than relying on a single probe, improves stability. This thesis adopts MLM probing as the primary naming backend and proposes a diversity-aware exemplar selection strategy to improve the quality of aggregate predictions.

2.0.7 Parameter-Efficient Fine-Tuning and Soft Prompts

Parameter-efficient fine-tuning methods reduce the number of trainable parameters while maintaining competitive performance. Prefix-tuning [28] and P-tuning [29] prepend trainable continuous tokens to the input while keeping

the backbone frozen. The Power of Scale work [30] shows that prompt tuning becomes competitive with full fine-tuning at larger model scales. LoRA [31] injects low-rank update matrices into attention layers. These methods are particularly relevant in open-world NER where training data from the source domain may be limited, motivating the soft prompt variant introduced in this thesis.

2.0.8 Large Language Models for Cluster Labeling

The emergence of large language models [32] has opened a new avenue for cluster labeling: instead of querying a masked language model, one can prompt a generative model with a set of representative cluster examples and ask it to produce a category name. Systems such as UniversalNER [18] and GOLLiE [33] demonstrate the versatility of LLMs for open-vocabulary information extraction. The main practical challenge is selecting which examples to include in the prompt: redundant examples waste context and introduce noise, while centroid-only selection may miss the full semantic range of the cluster. This thesis addresses this with Maximal Marginal Relevance [5] for exemplar selection.

2.0.9 Maximal Marginal Relevance

Maximal Marginal Relevance was originally proposed for query-based document summarization and result set diversification in information retrieval [5]. MMR iteratively selects items that are both relevant to a query and dissimilar

to already-selected items, controlled by a trade-off parameter λ . It has since been applied to sentence selection in extractive summarization, example selection for in-context learning, and retrieval-augmented generation. In this thesis, MMR selects a diverse subset of entity mention sentences from each cluster, with the cluster centroid serving as the query, ensuring the naming model receives evidence spanning the cluster’s full semantic range.

2.0.10 Positioning of This Work

This thesis does not propose a new open-world NER framework from scratch. Instead, it takes the OWNER pipeline as the base system and improves two components: the span detection stage is upgraded from a BIO-only head to the multi-view DTrans detector [4], and the cluster naming stage is enhanced with MMR-based diverse exemplar selection and a soft prompt entity typing variant. Compared to the original OWNER naming procedure, which uses a single nearest-centroid template per cluster, the proposed approach produces more specific, stable, and interpretable type names across all five CrossNER [6] target domains.

Chapter 3

Methodology

3.1 System Overview

Span2Type is a three-stage pipeline for open-world NER that takes an unannotated target-domain corpus as input and produces a set of entity spans, each assigned to a human-readable entity type. The three stages are:

1. **Entity Detection:** Given a sentence, detect all entity mention spans using a multi-view span detector trained on a labeled source domain and transferred to the target domain without retraining.
2. **Entity Typing:** Encode each detected span using a fill-in-the-blank template and a pretrained masked language model, then cluster the resulting embeddings to assign unsupervised type identifiers. A soft prompt variant replaces full fine-tuning with parameter-efficient prompt tuning.
3. **Automatic Cluster Naming:** Assign a concise, human-readable name to each cluster by selecting diverse representative exemplars via Maximal Marginal Relevance and querying either an MLM or a large language model for consensus type predictions.

The pipeline assumes no target-domain labels at any stage. Stages 2 and 3 are applied entirely to the target domain. Stage 1 uses source-domain labels only for training the entity detector. Table 3.1 summarizes the notation used throughout this chapter.

Table 3.1: Notation used in this chapter.

Symbol	Description
$\mathbf{x} = (x_1, \dots, x_T)$	Input token sequence of length T
$\mathcal{S}$	Set of detected entity spans in a sentence
$s = (i, j)$	Entity span from token i to token j (inclusive)
$\mathbf{h}_t$	Contextual hidden state of token t from the encoder
$\mathbf{e}_s$	Embedding of entity span s at the [MASK] position
k	Number of entity type clusters
k^*	Optimal k selected by BIC
$c_s \in \{1, \dots, k\}$	Cluster assignment of span s
μ_c	Centroid of cluster c
$\mathcal{E}_c$	Set of entity spans assigned to cluster c

3.2 Stage 1: Multi-View Entity Detection

3.2.1 Motivation

The original OWNER pipeline uses a DeBERTa-v3-base encoder with a single BIO sequence labeling head to detect entity spans. While BIO tagging is effective when the source and target domains are similar, it captures only one view of span boundary information. An entity mention can be character-

ized from at least three complementary perspectives: its internal BIO label sequence, the positions of its start and end tokens, and the pairwise “tied or broken” relationship between adjacent tokens. Using all three simultaneously produces richer and more transferable span representations.

This thesis replaces OWNER’s BIO-only detector with the Entity Detection (ED) module from the Divide-and-Transfer (DTrans) paradigm [4], which implements exactly this multi-view strategy.

3.2.2 Architecture

The ED module uses BERT-base-cased [10] as its backbone encoder. Given an input sequence $\mathbf{x}$, the encoder produces contextual representations $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_T) \in \mathbb{R}^{T \times d}$. Three independent classification heads are applied to $\mathbf{H}$ in parallel:

BIO decoder. A linear classifier maps each token hidden state to a three-class distribution over $\{B, I, O\}$:

$$p_t^{\text{BIO}} = \text{softmax}(\mathbf{W}_{\text{bio}} \mathbf{h}_t + \mathbf{b}_{\text{bio}}), \quad (3.1)$$

where $\mathbf{W}_{\text{bio}} \in \mathbb{R}^{3 \times d}$. This head captures the global sequential context of entity spans.

Start-End (SE) decoder. Two binary classifiers independently predict whether each token is the start or end of an entity span. The end classifier

conditions on the predicted start probability to improve boundary coherence:

$$p_t^{\text{start}} = \text{softmax}(\mathbf{W}_{\text{start}} \mathbf{h}_t), \quad p_t^{\text{end}} = \text{softmax}(\mathbf{W}_{\text{end}} [\mathbf{h}_t; \hat{p}_t^{\text{start}}]), \quad (3.2)$$

where $[\cdot; \cdot]$ denotes concatenation and $\hat{p}_t^{\text{start}}$ is the soft start label (one-hot during training, softmax output during inference). This head provides explicit boundary position signals that complement the sequential BIO labels.

Tie-Break (TB) decoder. A binary classifier predicts whether each adjacent pair of tokens $(t, t+1)$ belongs to the same entity span (*Tie*) or crosses a boundary (*Break*). The hidden representation for the pair is computed as the element-wise sum of the hidden states of both tokens:

$$\mathbf{h}_t^{\text{pair}} = \mathbf{h}_t + \mathbf{h}_{t+1}, \quad p_t^{\text{tb}} = \text{softmax}(\mathbf{W}_{\text{tb}} \mathbf{h}_t^{\text{pair}}). \quad (3.3)$$

This head explicitly models token-level adjacency within spans, capturing information that BIO and SE alone may miss.

3.2.3 Training Objective

The model is trained on the labeled source domain. The total loss is the sum of the cross-entropy losses from all three decoders:

$$\mathcal{L}_{\text{ED}} = \mathcal{L}_{\text{BIO}} + \mathcal{L}_{\text{SE}} + \mathcal{L}_{\text{TB}}, \quad (3.4)$$

where $\mathcal{L}_{\text{SE}} = (\mathcal{L}_{\text{start}} + \mathcal{L}_{\text{end}})/2$.

To stabilize training and improve generalization, a **Mean Teaching** strategy based on Exponential Moving Average (EMA) is applied. A teacher

model $\tilde{\theta}$ maintains a running average of the student model parameters θ :

$$\tilde{\theta}_t = \alpha \tilde{\theta}_{t-1} + (1 - \alpha) \theta_t, \quad (3.5)$$

with $\alpha = 0.995$. The teacher generates soft pseudo-labels on unlabeled data, which are added as an auxiliary consistency loss weighted by λ_w , encouraging the student to produce stable predictions.

3.2.4 Inference

At inference time, the three decoders independently decode span candidates from their respective predictions. The final entity span set is the union of candidates from all three decoders after applying a confidence threshold δ to filter low-confidence predictions. Spans from different views that overlap are merged, giving the downstream entity typing stage a high-recall set of candidates.

3.3 Stage 2: Entity Typing

3.3.1 Fill-in-the-Blank Entity Representation

For each detected entity span $s = (i, j)$ in sentence $\mathbf{x}$, the entity typing stage constructs a type-sensitive representation by filling a template. The template used throughout this thesis is:

$$\text{template}(s, \mathbf{x}) = [\text{CLS}] \ x_1 \cdots x_T \ m \text{ is a } [\text{MASK}] \ . \ [\text{SEP}], \quad (3.6)$$

where $m = x_i \cdots x_j$ is the surface form of the entity mention. This template is tokenized and passed through a BERT-based encoder. The hidden state at the [MASK] position serves as the entity embedding $\mathbf{e}_s \in \mathbb{R}^d$:

$$\mathbf{e}_s = \text{Dropout}(\mathbf{h}_{[\text{MASK}]}) , \quad (3.7)$$

where $\mathbf{h}_{[\text{MASK}]}$ is the last-layer hidden state at the mask position. Because the template implicitly asks the model to predict the type category of the entity, the [MASK]-position representation is more type-discriminative than a raw span embedding.

3.3.2 Training with Batch Triplet Margin Loss

The entity encoder is fine-tuned on the source-domain training set using a **Batch Triplet Margin Loss** [34]. For a batch of N entities with embeddings $\{\mathbf{e}_i\}_{i=1}^N$ and source-domain type labels $\{y_i\}_{i=1}^N$, all valid (*anchor, positive, negative*) triplets are extracted in-batch. A triplet (i, j, k) is valid if $y_i = y_j$ and $y_i \neq y_k$. The loss is:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} \max(0, \|\mathbf{e}_i - \mathbf{e}_j\|_2 - \|\mathbf{e}_i - \mathbf{e}_k\|_2 + \gamma) , \quad (3.8)$$

where $\mathcal{T}$ is the set of valid triplets and $\gamma = 1.0$ is the margin. This objective pushes embeddings of the same type closer together and embeddings of different types further apart, yielding a metric space where k -means clustering recovers semantically coherent entity type groups.

3.3.3 Soft Prompt Variant

Standard fine-tuning updates all parameters of the BERT encoder, which can be expensive and may overfit when training data is limited. This thesis also proposes a **soft prompt** variant inspired by P-tuning [29,30], where the base encoder is kept frozen and a small set of N_p trainable soft token embeddings $\mathbf{P} = [\mathbf{p}_1; \dots; \mathbf{p}_{N_p}] \in \mathbb{R}^{N_p \times d}$ is prepended to the token embedding sequence before being passed to the frozen encoder:

$$\text{inputs_embeds} = [\mathbf{P}; \mathbf{E}_x] \in \mathbb{R}^{(N_p+L) \times d}, \quad (3.9)$$

where $\mathbf{E}_x$ are the standard token embeddings of the template-expanded input of length L . The attention mask is extended accordingly to cover the prepended soft tokens. The [MASK]-position index is shifted by N_p to account for the prepended tokens. Only $\mathbf{P}$ (and optionally the encoder with a lower learning rate) is updated during training; the backbone remains frozen. With $N_p = 5$ (default), this reduces the number of trainable parameters from $\sim 110\text{M}$ to just $5 \times 768 = 3,840$, making the method significantly more parameter-efficient.

The same batch triplet margin loss (Equation 3.8) is used to train the soft prompt parameters.

3.4 Stage 3: BIC-Guided Automatic Cluster Assignment

3.4.1 AutoKmeans

After the encoder is trained, all entity spans in the target domain are encoded into embeddings $\{\mathbf{e}_s\}$. Since the number of entity types in the target domain is unknown, the system must determine k automatically. This is achieved via **AutoKmeans**, which searches over a range $[k_{\min}, k_{\max}]$ with step k_{step} and selects the k^* that minimizes the Bayesian Information Criterion (BIC).

For a fitted k -means model [24] with inertia $\mathcal{I}(k)$ (sum of squared distances to cluster centroids) on n samples, the BIC [26] is:

$$\text{BIC}(k) = n \cdot \log\left(\frac{\mathcal{I}(k)}{n}\right) + \log(n) \cdot k. \quad (3.10)$$

The first term measures fit quality: it decreases as k grows and the clustering better approximates the data. The second term is a complexity penalty that increases with k . The optimal number of clusters is:

$$k^* = \arg \min_{k \in \{k_{\min}, k_{\min} + k_{\text{step}}, \dots, k_{\max}\}} \text{BIC}(k). \quad (3.11)$$

GridSearchCV with k -means++ initialization and 10 random restarts is used for each candidate k . The best model (at k^*) is then used to assign cluster labels $c_s \in \{1, \dots, k^*\}$ to every entity span.

3.5 Stage 4: Automatic Cluster Naming

3.5.1 MMR-Based Exemplar Selection

Given a cluster c with entity spans $\mathcal{E}_c$ and their embeddings, the naming module first selects a small set of K_e diverse, centroid-relevant exemplar sentences using **Maximal Marginal Relevance** [5].

Let $\boldsymbol{\mu}_c = \frac{1}{|\mathcal{E}_c|} \sum_{s \in \mathcal{E}_c} \mathbf{e}_s$ be the cluster centroid. MMR iteratively builds the exemplar set $\mathcal{R}$ by selecting at each step the span that maximizes:

$$\text{MMR}(s) = \lambda \cdot \cos(\mathbf{e}_s, \boldsymbol{\mu}_c) - (1 - \lambda) \cdot \max_{r \in \mathcal{R}} \cos(\mathbf{e}_s, \mathbf{e}_r), \quad (3.12)$$

where $\lambda \in [0, 1]$ controls the trade-off between centroid relevance (first term) and diversity relative to already-selected exemplars (second term). A value of $\lambda = 0.7$ is used by default, slightly favouring relevance over diversity. The process repeats until $|\mathcal{R}| = K_e$ (default $K_e = 5$). This ensures that the selected exemplars cover the semantic spread of the cluster rather than all clustering near the centroid.

3.5.2 MLM-Based Naming

In the MLM backend, each selected exemplar sentence $x^{(r)}$ containing entity mention $m^{(r)}$ is inserted into a naming template, e.g.:

In this sentence, $m^{(r)}$ is a [MASK]. Sentence: $x^{(r)}$.

The template is tokenized and passed through a pretrained BERT MLM. The probability distribution over the vocabulary at the [MASK] position is

extracted, and the distributions from all K_e exemplars are averaged:

$$\bar{p}_v = \frac{1}{K_e} \sum_{r=1}^{K_e} p_v^{(r)}, \quad v \in \mathcal{V}. \quad (3.13)$$

The candidate tokens are filtered to remove sub-word pieces, special tokens, punctuation, and common stopwords. The top-ranked remaining token is selected as the cluster name. This consensus over diverse exemplars produces more stable and specific names than querying a single representative instance.

3.5.3 LLM-Based Naming

In the LLM backend, the K_e MMR-selected exemplars are formatted as a numbered list and presented to a locally hosted large language model (via Ollama) with the prompt:

You are naming a semantic entity cluster. Infer one shared category from the examples.

Rules: Output ONLY one category name. Use 1 to 3 words. No explanation, no punctuation.

Examples: [numbered list of exemplars]

Category:

The model response is post-processed to extract at most three words, strip filler phrases, and normalize to lowercase. The LLM backend can produce multi-word type names (e.g., RESEARCH INSTITUTE) that single-token MLM predictions cannot, at the cost of requiring a locally available language model.

The two naming backends are interchangeable within the pipeline; ablation experiments in Chapter 4 compare their naming quality.



Chapter 4

Experiments and Results

4.1 Experimental Setup

4.1.1 Datasets

We evaluate Span2Type in a *cross-domain open-world* setting. The entity detector is trained on a labeled source domain and then applied to unseen target domains; no target-domain labels are used at any stage except for evaluation.

Source datasets. We use two source corpora to train the DTrans entity detector:

- **CoNLL-2003** [1]: English newswire with four entity types (PER, ORG, LOC, MISC). This is the standard source corpus used in cross-domain NER benchmarks.
- **Pile-NER**: A large, diverse NER corpus derived from the Pile, covering a broad range of entity categories. Training on Pile-NER exposes the detector to a wider variety of entity surface forms.

Target datasets. We evaluate on five target domains from the **Cross-NER** [6] benchmark, a standard cross-domain evaluation suite with gold entity annotations:

- **AI:** Scientific text about artificial intelligence (14 entity types).
- **Literature:** Text about literary works and authors (12 entity types).
- **Music:** Text about music and musicians (13 entity types).
- **Politics:** Text about political figures and events (9 entity types).
- **Science:** Text about natural sciences (17 entity types).

For all target domains, gold labels are used strictly for evaluation; they are never seen during training or clustering.

4.1.2 Implementation Details

Entity Detection. The DTrans multi-view entity detector uses **BERT-base-cased** [10] as the backbone encoder (12 layers, hidden size 768). Training hyperparameters follow [4]: learning rate 1×10^{-5} , batch size 32, 50 epochs, warmup steps 500, EMA decay $\alpha = 0.995$, consistency weight 100. The maximum sequence length is 128 tokens. Experiments are run with two GPUs.

Entity Typing. The entity encoding model uses **BERT-base-uncased** with the template “*{sentence} {entity} is a [MASK].*”. The model is fine-tuned

with AdamW (learning rate 2×10^{-5}) and a batch triplet margin loss ($\gamma = 1.0$) for up to 30 epochs with a linear warmup scheduler. For the soft prompt variant, the base BERT is frozen and $N_p = 5$ trainable soft tokens are prepended; separate learning rates are used for soft prompt parameters and PLM parameters.

Clustering. AutoKmeans searches $k \in [2, 30]$ with step 2, using k -means++ initialization and 10 random restarts per k value.

Cluster Naming. MMR exemplar selection uses $K_e = 5$ exemplars with $\lambda = 0.7$. The MLM backend uses BERT-base-uncased. The LLM backend uses Llama-3 served via Ollama.

Reproducibility. All experiments use a fixed random seed (0). Code and configuration files are included with the submission.

4.2 Evaluation Metrics

We report metrics at two levels of the pipeline.

Entity Detection. We report span-level **Precision**, **Recall**, and **F1** on the test set of each target domain, using exact span match. This evaluates whether the multi-view DTrans detector transfers better than the OWNER BIO-only baseline.

Entity Typing. We report **Hungarian-matched Micro-F1** and **Macro-F1** to measure overall and per-type typing accuracy after optimal cluster-to-type assignment. We also report **Adjusted Mutual Information (AMI)** [27], a mapping-free metric that measures clustering quality without requiring a bipartite assignment.

Cluster Naming. We evaluate predicted cluster names against gold entity type names using **BERTScore-F1** [35], which measures semantic similarity between predicted and reference names using contextual embeddings.

4.3 Baselines and Ablations

We compare against the following systems:

OWNER (baseline). The original OWNER pipeline [3] uses a DeBERTa-v3-base BIO mention detector trained on the source domain, BERT fill-in-the-blank entity embeddings, BIC-guided k -means clustering, and single-template MLM naming without exemplar selection.

Span2Type-BIO. Replaces the DTrans multi-view detector with a single BIO head (same BERT backbone), keeping all other components identical to Span2Type-Full. This ablation isolates the contribution of multi-view detection.

Span2Type-SoftPrompt. Replaces the full BERT fine-tuning in the entity typing stage with the soft prompt variant (frozen PLM, $N_p = 5$ trainable tokens). All other components are identical to Span2Type-Full.

Span2Type-CentroidNaming. Uses the centroid-nearest single exemplar for cluster naming instead of MMR-based diverse selection. This ablation isolates the contribution of exemplar diversity.

Span2Type-Full. The complete system: DTrans multi-view entity detection + full BERT entity typing + BIC k -means + MMR-based MLM naming.

4.4 Entity Detection Results

Table 4.1 compares span detection F1 across five CrossNER target domains for different mention detection architectures, all trained on CoNLL-2003.

Table 4.1: Mention detection evaluation results for different architectures trained on CoNLL and tested on five $\mathcal{D}^T$ datasets. Performance is measured using the F1 score. Best results are in bold.

Model	AI	Literature	Music	Politics	Science	Average
BIO (OWNER)	60.5 (4)	83.9 (5)	79.0 (2)	80.6 (2)	74.4 (7)	75.68
PURE	39.8	37.1	33.8	32.4	35.7	35.76
SpanProto	54.1	62.9	59.6	68.7	59.7	61.00
WL-Coref	57.4 (8)	68.3 (1.7)	66.9 (2.1)	72.1 (1.3)	63.4 (3.3)	65.62
BIO-SE-TB (ICC)	81.35	82.68	82.68	89.64	75.41	82.35

The DTrans multi-view model (BIO-SE-TB) achieves the highest average F1 of 82.35, outperforming the OWNER BIO baseline (75.68) by 6.67 points on average. The gains are consistent across all five domains, with the most notable improvement in AI (+20.85 points over OWNER BIO). The competing span detection methods PURE (35.76) and SpanProto (61.00) lag significantly behind, while WL-Coref (65.62) performs better but still falls short of the multi-view approach. Literature is the only domain where OWNER BIO achieves the top score (83.9), narrowly surpassing BIO-SE-TB (82.68), likely because the literary domain’s entity boundaries align well with the sequential BIO formulation. Overall, the results confirm that jointly training three complementary decoders yields substantially more robust span detection across domains.

4.5 Main Results

Table 4.2 reports end-to-end NER F1 on the five CrossNER target domains, comparing Span2Type against unsupervised and open-world NER baselines. All methods use CoNLL-2003 as the source unless stated otherwise.



Table 4.2: Comparison of different Named Entity Recognition (NER) methods across various domains. Best results are in bold.

Method	AI	Liter.	Music	Politics	Science	Avg.
ChatIE Uns	12.8	20.4	27.8	27.5	25.7	22.8
ChatIE Uns (GPT-4o mini)	21.8	37.5	43.7	37.2	35.0	35.0
ChatIE Uns (Llama 3.1)	17.9	29.2	35.6	32.1	29.4	28.8
UniNER Uns	33.5	42.8	48.1	40.3	43.7	41.7
UniNER Uns (GPT-4o mini)	36.7	49.8	53.6	48.8	48.1	47.4
OWNER (CoNLL)	44.3 (3)	46.6 (5)	50.1 (9)	53.7 (3)	52.3 (6)	49.4
OWNER (Pile-NER)	39.4 (9)	49.5 (8)	52.5 (3)	48.5 (7)	50.9 (4)	48.2
Span2Type (CoNLL)	57.6	56.9	60.0	66.3	66.0	61.4
Span2Type (Pile-NER)	50.0	52.8	56.16	55.84	62.29	55.42

Span2Type (CoNLL) achieves the best average F1 of 61.4, outperforming the next best system OWNER (CoNLL) by 12.0 points (49.4 $\rightarrow$ 61.4). The improvements are consistent across all five domains: AI (+13.3), Literature (+10.3), Music (+9.9), Politics (+12.6), and Science (+13.7). Compared to OWNER trained on Pile-NER (48.2), Span2Type (CoNLL) still gains 13.2 points on average.

Among LLM-based unsupervised baselines, UniNER with GPT-4o mini (47.4) is the strongest, but still falls 14.0 points below Span2Type (CoNLL). The ChatIE-based approaches lag considerably at 22.8–35.0, demonstrating that open-world NER requires more than zero-shot prompting.

Span2Type (Pile-NER) achieves 55.42 on average, which is lower than the CoNLL-source variant (61.4) but still substantially above all baselines.

This suggests that the broader vocabulary coverage of Pile-NER does not consistently benefit the CrossNER domains, which are closer in style to the structured CoNLL-2003 source.

4.6 Cluster Naming Results

Table 4.3 reports cluster naming quality measured by BERTScore-F1 between predicted cluster names and gold CrossNER entity type names.

Table 4.3: Cluster naming quality (BERTScore-F1) averaged over the five CrossNER target domains. Higher is better. **Bold** indicates best result.

Method	AI	Lit.	Music	Pol.	Sci.	Avg.
OWNER (single-template MLM)	—	—	—	—	—	—
Span2Type-CentroidNaming (MLM)	—	—	—	—	—	—
Span2Type-MLM (MMR)	—	—	—	—	—	—
Span2Type-LLM (MMR)	—	—	—	—	—	—

4.7 Ablation Study

Table 4.4 isolates the contribution of each Span2Type component by progressively removing or replacing one part of the full system.

Table 4.4: Ablation study of Span2Type components across different domains. Performance is measured using F1 score. Best results are in bold.

Method	AI	Liter.	Music	Politics	Science	Avg.
OWNER (CoNLL) w/o all components	44.3	46.6	50.1	53.7	52.3	49.4
Span2Type w/o Soft Prompt	52.3	56.3	57.5	63.9	62.8	58.6
Span2Type w/o Ensemble MD	49.31	53.01	58.84	64.09	63.75	57.80
Span2Type (Full Model)	57.6	56.9	60.0	66.0	66.3	61.4

The OWNER baseline (49.4 avg) serves as the starting point. Replacing the BIO-only mention detector with the DTrans multi-view ensemble while keeping all other components identical (“w/o Soft Prompt”) improves the average F1 to 58.6 (+9.2 points), demonstrating that multi-view entity detection is the single largest contributor to the overall improvement.

Adding the soft prompt entity typing variant on top brings the full model to 61.4 (+2.8 over “w/o Soft Prompt”), showing that parameter-efficient prompt tuning further refines the entity type embeddings even with the backbone frozen. Removing the ensemble mention detector while keeping soft prompt tuning (“w/o Ensemble MD”, 57.80) shows a 3.6-point drop relative to the full model, confirming that both components contribute independently.

These results indicate that the three improvements are complementary: the DTrans multi-view detector provides higher-quality span candidates, the soft prompt tuning produces more type-discriminative embeddings, and together they push the full pipeline to 61.4 average F1.

4.8 Discussion

Effect of multi-view entity detection. Table 4.1 shows that the DTrans multi-view detector (BIO-SE-TB) achieves an average F1 of 82.35 across five CrossNER domains, compared to 75.68 for the OWNER BIO-only baseline—a gain of 6.67 points. The improvement is largest in AI (+20.85) and smallest in Literature, where the BIO baseline remains competitive (83.9 vs 82.68). This suggests that the complementary SE and TB decoders are most beneficial in domains with irregular entity boundaries and diverse surface forms, while sequential BIO labeling suffices for more structured literary text.

End-to-end NER improvement. The main results in Table 4.2 show that Span2Type (CoNLL) achieves 61.4 average F1, a 12.0-point improvement over OWNER (49.4). This gap is substantially larger than the 6.67-point improvement at the entity detection stage alone, suggesting that the better span candidates from DTrans detection also improve downstream entity typing quality by reducing noise in the embeddings used for clustering.

Effect of source corpus. Span2Type (Pile-NER) achieves 55.42 average F1, lower than the CoNLL-source variant (61.4). The CrossNER domains are stylistically closer to CoNLL-2003 newswire than to the heterogeneous Pile-NER distribution, so the more focused CoNLL source produces a better-calibrated detector for these benchmarks.

Soft prompt tuning and ensemble detection. The ablation in Table 4.4 confirms that both the multi-view detector (+9.2 from 49.4 to 58.6) and the soft prompt entity typing (+2.8 from 58.6 to 61.4) contribute independently. The soft prompt variant reduces trainable parameters from $\sim 110\text{M}$ to $\sim 3,840$ while still improving over the full fine-tuning OWNER baseline, demonstrating a favorable efficiency–performance trade-off.



Chapter 5

Conclusion

This thesis presented **Span2Type**, an improved open-world NER pipeline built on the OWNER framework, with three targeted enhancements targeting the two weakest points of the original system: span detection quality and automatic entity type naming.

5.1 Summary of Contributions

We introduced three improvements to the OWNER pipeline and evaluated them systematically on the CrossNER benchmark across five target domains.

First, we replaced OWNER’s single BIO mention detector with the DTrans multi-view Entity Detection module [4], which jointly trains BIO, Start-End, and Tie-Break decoders on the source domain with a Mean Teaching strategy. This change alone improved span detection F1 from an average of 75.68 (OWNER BIO) to 82.35 (BIO-SE-TB), a gain of 6.67 points across five CrossNER domains.

Second, we introduced a soft prompt variant of the entity typing stage in which only a small set of $N_p = 5$ trainable soft tokens is optimized while the BERT backbone remains frozen. This reduces trainable parameters from $\sim 110\text{M}$ to $\sim 3,840$ while still improving end-to-end NER performance, of-

fering a parameter-efficient alternative to full fine-tuning that is attractive in memory-constrained or limited-data settings.

Third, we proposed an MMR-based exemplar selection strategy for automatic cluster naming. Rather than querying a naming model with the single centroid-nearest example per cluster, we use Maximal Marginal Relevance [5] to select $K_e = 5$ diverse, centroid-relevant exemplars before passing them to either an MLM or a locally hosted LLM. This ensures the naming model receives evidence spanning the full semantic range of the cluster, producing more specific and stable type names.

On the CrossNER benchmark, Span2Type (CoNLL source) achieves an average end-to-end NER F1 of **61.4**, compared to **49.4** for the OWNER baseline—a 12.0-point improvement. Ablation experiments confirm that the multi-view detector contributes the largest share of this gain (+9.2 points), while soft prompt tuning provides a complementary additional improvement (+2.8 points).

5.2 Limitations

Despite these improvements, Span2Type retains several limitations inherited from the OWNER framework.

The entity detector is still trained on a labeled source domain and transferred to target domains; in settings where the source and target domains are very distant (e.g., newswire to clinical text), transfer quality degrades. Future work could explore self-training or domain-adaptive pretraining to bridge larger domain gaps.

The BIC-guided k selection works well when the true number of entity types falls within the search range $[k_{\min}, k_{\max}]$, but it can still misestimate k for domains with highly imbalanced type distributions or very fine-grained entity inventories. A more principled nonparametric clustering approach could improve robustness here.

The cluster naming quality depends on the availability and quality of the MLM or LLM backend. Single-token MLM names are sometimes too generic (e.g., *entity*, *person*), while LLM-based names, though richer, require a locally hosted model that may not always be accessible. Improving the post-processing and filtering of generated names remains an open problem.

5.3 Future Work

Several directions extend the work presented in this thesis. First, the soft prompt tuning approach could be extended with adapter layers or LoRA-based fine-tuning to capture more expressive adaptation with a still-limited parameter budget. Second, the MMR exemplar selection strategy could be applied not only at naming time but also during the entity typing training phase, selecting diverse in-context examples for few-shot or retrieval-augmented typing. Third, the BIC-guided clustering could be replaced with a hierarchical or density-based method that produces a type hierarchy rather than a flat partition, enabling more nuanced ontology induction. Finally, Span2Type could be evaluated on additional benchmarks beyond CrossNER, particularly in biomedical and legal domains where the need for open-world entity type discovery is most acute.

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